Efficient Food Waste Detection and Classification in Real-time with Edge Computing

Abstract— **In today's world, the risk of a rising food crisis increases every day. This has made it necessary to take remediation steps. Steps towards creating sustainable methods for preserving food and reducing food wastage. Mess halls, canteens, and dining events are the largest sources of food wastage. This is because of the large number of guests that they entertain. Thus, such places need a solution for waste reduction. In this research, we tackle this problem by implementing a machine-learning solution. We pair this with computer vision to input image data straight from the waste bin. This method not only detects but also classifies the wasted food items. The proposed food detection and segregation system works by using computer vision. It then captures various dishes and food items within a used plate. After this, it identifies the wasted food items and how much-wasted food was on a plate. Along with this, the weight data is collected from the Load Cells placed below the collection bins. The data collected over several days can then be analyzed. This analysis can reveal the problematic dishes that are being wasted. Such dishes can then either be stricken from the menu or improved in quality.**

Keywords—food wastage, machine learning, computer vision, IoT

I. Introduction

In scientific literature, the terms food loss and food waste are commonly employed to describe items meant for human consumption that are later discarded, lost, degraded, or contaminated. Globally, approximately 25% to 33% of food is discarded, resulting in 1.3 × 109 tons of food waste each year [1]. Concerns about the security of food supplies brought on by the rise of the population are drawing attention to the Sustainable Development Goals (SDGs). This emphasizes the importance of the circular economy idea, which aims to minimize waste and maximize resources. A significant amount of food waste leads to social and environmental concerns. This includes challenges with social structure, overuse of land, economic hardships, food security, the greenhouse effect, and unequal distribution of food worldwide.[2]

This paper follows the given format. In section 2 we discuss the various other works that have been performed on data imputation and missing value prediction in real-time and near-streaming environments. Section 3 goes into detail about our proposed method, each of its components, and their respective implementation, details, and advantages. The experimentation, its details, specific parametric values, the details of the database, and the evaluation metrics are covered in section 4. Results and discussion related to the experimentation and the model take place in section 5, and finally, section 6 covers the conclusion and future work related to our approach. Additionally, section 7 is used to show the various scientific and technological studies we referenced in this work.

II. Related works

Food wastage monitoring has been a topic under research for a long time. Recently advances in the fields of IoT and the easy availability of sensors have given rise to much more such efforts. Gull S. et al in their work with using IoT for food wastage monitoring, took the approach of using various gas sensors.[3] This allowed them to train a decision tree-based machine learning algorithm on the proportions of the various gases emitted by the different food items. However, this system only works once the food goes stale and starts to rot. Only when a sufficient amount of oxidization occurs are the gases emitted from the food for the detection to work.

Food wastage is a problem that affects the entire process and pipeline of food preparation from farms to shipping, processing, packaging, distribution, preparation at home, and consumption. Chauhan Y. discusses how various technological advancements like automated PLCs and ERP can help reduce food wastage during the various shipping and logistical processes of pipelines.[4] While these methods and technological advancements are advantageous in some places they do not cover the reduction and management of food wastage at the ultimate destination of the consumer.

Universities and their cafeterias often become the source of large amounts of food waste even in smaller communities and towns. This food wastage sources from the large volume of food to be served and prepared in a short time and the variety of services these cafeterias can provide ranging from pre-prepared meals to custom order dishes. Faezirad M. et al proposed a neural network-based demand prediction model.[5] This model predicts the demand for food dishes on various days by the students and thus adjusts the amount of food prepared as per the demand. However, the training and implementation of such a model heavily depends on a food reservation and booking system being in place at the university which is not the case for many smaller universities, colleges, and the cafeterias within.

Alternate solutions target major food wastage sources such as hotels to better manage their food and produce less waste. Kumar U. et al talk about the use of mobile-based applications to track food wastage and connect such hotels to NGOs and Social groups who can later distribute the leftover food to the needy.[6] Bharucha J. in their work talks about the various ways in which restaurants deal with surplus food including giving away food to the staff, NGOs/Orphans, Slums, etc.[7] While these methods look at the social welfare aspect of food management, the method requires a high amount of active participation from multiple parties.

The other problem with such methods which involve the donation of food after consumption by the primary consumer is the effort and cost of hygienically storing the food, transporting the already cooked food to various social welfare sites, and matching the type of food to the preferred secondary consumer which can be complicated due to cultural dietary constraints. Filimonau V. in their research found various restaurants preferred actively working towards making sure no extra food or big portions are prepared rather than alternative disposal methods such as donating.[8]

Various government schemes that work to supply food for welfare also result in wastage of food due to mismanagement of funds, poor quality of food, loss of produce during transportation, and revision of shelf-life policies. With better preservatives and storage conditions in modern facilities, the shelf life of food produce can be widely increased compared to

when the policies mandating their shelf life were first thought of, such policies need to be revisited and updated with the current up-to-date technologies. Saadat A. et al talk about schemes such as "Akshaya Patra (Aganwadi)".[9] While having goals such as providing food for the preschool non-formal education for children, providing scientifically attested nutrition and health education; promoting and enabling immunization; facilitating regular health check-ups; and providing the information and playing a role in referral services, end up, unfortunately, producing food wastage due to reasons such as corruption leading to low-quality food which is not plateable and thus not eaten and enjoyed by the children. The study further suggests the implementation of direct benefit transfer schemes which either give direct monetary funding to the families of such children for proper nutritional food budgeting or provide dry packets in bunches that can be cooked by the families for the children.

In their extensive research Sahoo A. et al talk about the various methods in which different countries deal with excessive food wastage.[10] This ranges from Incineration in Singapore and the USA to Animal feeding of food wastage in Japan and Korea. While these methods show how generated food can be dealt with such solutions do not target the actual generation of excess waste food.

III. Methodology

This section outlines the methodology employed in our research to detect and manage food wastage using edge computing, IoT, machine learning, and computer vision. The approach consists of several key stages: detection of the metal plate, identification of circular food items, and segmentation of the food plate into multiple sections for enhanced machine learning training.

A. System Architecture

The proposed system leverages edge computing to process data locally on IoT devices, minimizing latency and bandwidth usage. The architecture consists of the following components:

- **IoT Sensors**: Equipped with cameras to capture real-time images of food plates, load cells to capture the weight of the food plates, and ultrasonics sensors for management of the items in the garbage bin.
- **Edge Computing Device**: Processes images locally using machine learning algorithms, reducing the need for cloud-based processing.
- **Data Storage**: A cloud-based database to store segmented images and metadata for training and analysis.
- **Realtime Database:** A cloud-hosted real-time database to which the processed data can be sent to be accessed by applications and users for further use.

Our system was implemented on top of a rover-like bot to add additional functionality in the form of mobility to the system. In *Figure 1* the top view of the bot is shown. Along with this *Figure 2* shows the side view of the bot.

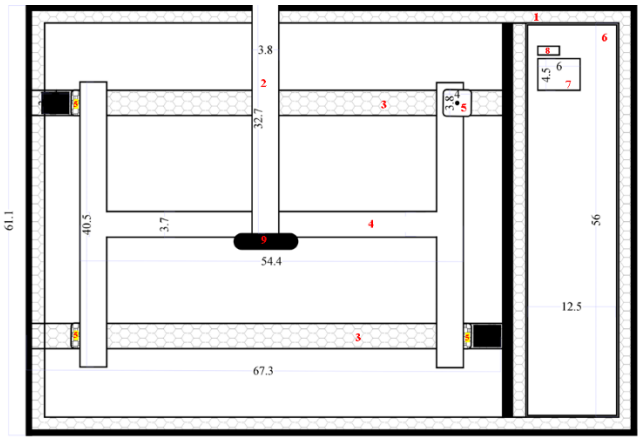


Figure 1 Top View of the System

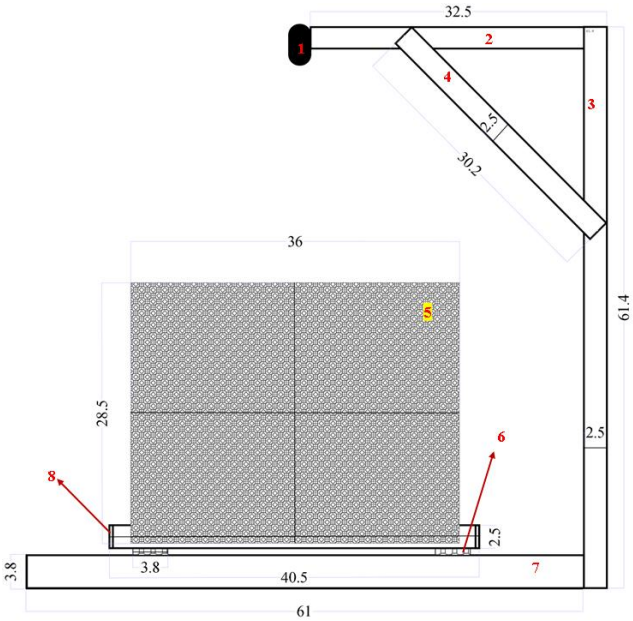


Figure 2 Side View of the System

Various numbers are present on the images which correspond with various important components labeled in Table 1.

Table 1 Various Parts of the Bot

S. No.	Component	Description
1.	Base Frame	The base frame is made out of aluminum and made with the high weight of multiple food plates in consideration.
2.	Camera Mount	Elevated and inverse L-shaped mounting for the camera to conveniently take pictures of the discarded food plate
3.	Cross Frame	Aluminum sections that hold the Load cells above which the food collection creates are positioned.
4.	Base for Crate	A special Base was created with aluminum plates to transfer the weight of the crate to the load cells evenly.
5.	Load Cell	50 Kgs Load cell with Amplifier module to capture the weight of the food plates through the crate.

6.	Base for Electronic Components	A base made out of board plastic is used to keep the various electrical components over.
7.	Micro Processor	Various Electrical components are necessary for the function of the system such as for edge computing and movement.
	Graphical Processor	
	Network Module	
	Motor Drivers	
8.	Amplifier Module	Amplification Module to be implemented in conjunction with the Load cells.
9	Camera Module	Standard webcam to take pictures of the food plates.

The bot is made using an aluminum frame to provide strength to handle multiple food plates and payload while also being very light. The bot features a vertically protruding rod with a bend at the top completing an L shape at the tip of which a camera is mounted directly at the center of the payload area to capture clear images of the food. At each corner of the payload area, 4 load cells each rated for 50 kgs of weight have been mounted using special 3d printed covers to input accurate weight data. A designated area with a base of insulative Sun board is being used to store all the electronics, computers, microcontrollers, and batteries being used by the bot.

The camera mount features multiple aligned holes on a lower and upper section with two nuts and bolts holding the upper and lower section together. These holes allow us to adjust the height of the camera module at any time and also ensure a stable mount for the camera. Various extra supports have been added to the mounting to make the camera stable even when the bot itself is moving around.

The bot is designed to support large crates which can be used as a bin for the food plates, these crates can be covered with large trash bags that will ensure no food/liquid is spilled affecting the bot or the surrounding area. The bot's modularity allows crates of varying sizes to be compatible with the bot, and no one crate needs to be quickly reused thus a smooth-flowing cycle with the bot and the cleaning staff can be maintained.

The algorithm that allows the bot to conduct its primary goal requires two pieces of input from the hardware, an image of the food plate, and its weight. These inputs are given to the algorithm by the load cells and the camera module attached to the bot, the four load cells at each corner of the payload area can measure a collective weight of 200 kg, and they are interfaced with a Microprocessor through an Amplifier chip, the Microprocessor is programmed to check the weight at a fixed interval and if the weight is sense passes a certain threshold (the weight of an empty plate), it sends a trigger signal along with the weight measurement to the Graphical Processor module to continue with the algorithm and process the information.

Once the processing is triggered a picture of the food plate is captured and sent to the computing unit for further processing, the camera module is a webcam that interfaces with the computing unit. The load cells and the camera unit communicate with different modules and thus need serial communication between the Microprocessor and the graphical computing unit.

Figure 2 is a basic schematic of the various hardware components and how they interact with each other. The load cells work on a wheat stone bridge principle, where the more the load the more resistance they produce. This resistance is read by an amplifier module and given to a microprocessor. The load if more than a specified threshold, triggers the process within the graphical processor, which takes a picture of the plate and analyses the food. This data along with the weight data is then sent to the database. The LEDs are used to convey the current start of the system to the user.

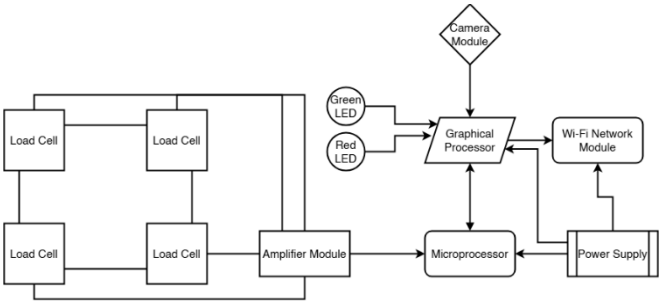


Figure 2 Schematic of the Electronic System

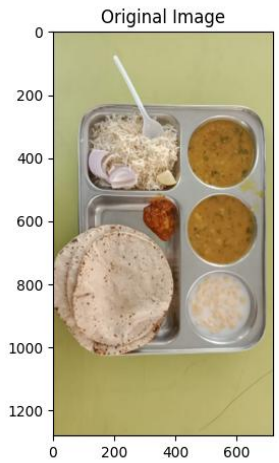
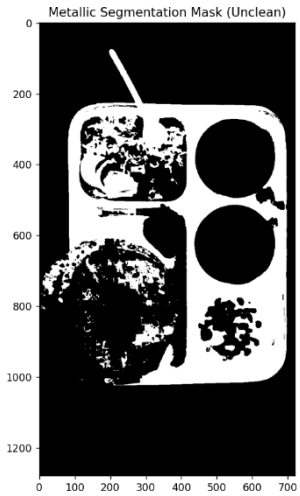
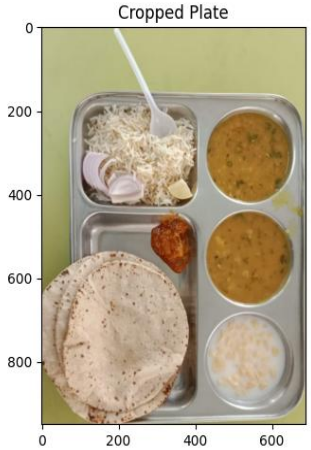
B. *Image Pre-Processing*

The images used while researching this model were collected by the team in real environments so that accurate conditions of lighting along with genuine wasted food in communal areas such as hostels, canteens, etc. can be researched. And so that different variations of food plates with varying configurations can be captured.

1) *Metal Plate Detection*

The first step in the methodology involves detecting the metal plate on which the food is served. The images are first converted to HSV color space from BGR. Upper and Lower bounds are supplied to define a range for metallic colors. The best results were obtained when using the upper bound of (220, 50, 225) and the lower bound of (0,0,70) both in HSV. Following this a mask is created and sent through morphological operations to fill in any irregular shapes or gaps in the mask. The mask is overlayed on top of the original image to define the boundaries that need to be cropped out. *Table 2* contains various stages of the Metal Plate Detection process showcasing the auto-cropping mechanism.

Table 2 Various Stages of Metal Plate Detection

I. 	II. 
In cell I The original image to be processed is shown.	In cell II the Image Segmentation Mask for the metallic plate is shown built using the provided upper and lower limits on HSV Values.
III.	IV. 
In cell III the Mask is again shown after going through morphological cleaning to achieve a better mask.	In cell IV the finally cropped image is showcased.

2) Food Item Detection

Once the metal plate is detected, the next step is to identify the food items on the plate. This is accomplished using the Hough Transform: The Hough Circle Transform is applied to detect circular shapes within the defined region of the metal plate. This method is used by us due to the plate having multiple round sections to hold the wet/gravy-based dishes.
The cropped image is first gray-scaled and then passed through a Gaussian blur filter to smooth it out. Following this, a Hough Circle algorithm is applied to obtain a list of all the detected circles. The inverse ratio for the accumulator used was 0.2. The minimum distance between the centers of the two detected circles was limited to 200 pixels and the upper and lower bounds of the radius of a circle are 75 and 175 pixels respectively. Finally, the upper threshold for the edge detector is 170 and the threshold for circle detection is 50.

The circles are checked to see if they contain any food items are not. As a section of a food plate can only have two possibilities, either presenting metal or presenting food, this detection can be achieved by iterating through all the detected circles and checking if the ratio of the metallic pixels to that of all pixels within the section is less than 0.9. Along with this, a second check is implemented using the Laplacian function which detects the rate of change of intensity of the image and acts as a texture identifier. If the segment has too much texture as opposed to a smooth flat surface, the section is likely to contain food.

3) Image Segmentation

After identifying the food items, the system segments the food plate into multiple sections. This is made easy by using the data from the previous steps by cropping out the detected circles that were detected to contain food within them. The detected circles are used as reference points to crop the original image into smaller sections. Each section corresponds to a specific food item or area of interest on the plate.
Figure 3 shows the various food items detected and cropped into individual images.

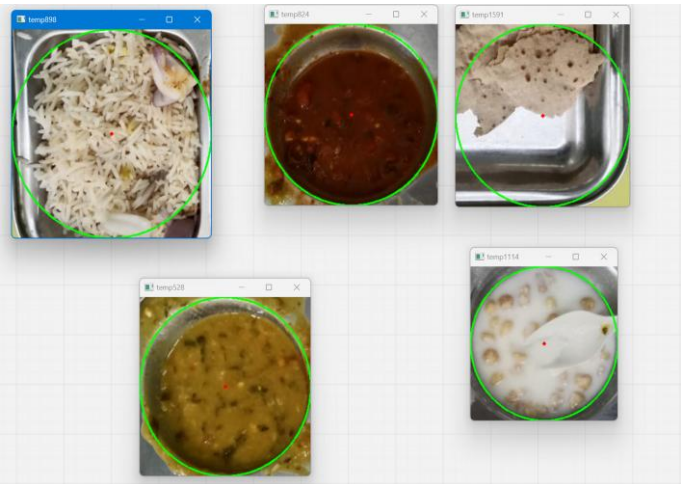


Figure 3 Cropping of Various Food Items

C. Model Training

Once it is confirmed that the circle contains food, it is sent to the CNN model to be classified. The base model used by us is a fine-tuned version of the Mobile Net V2 which is a pre-trained CNN with 53 layers. It has been trained on large datasets such as the ImageNet database.[11] The model is then taken through a transferred learning process using data collected in the institute canteen and hostels.

The image data acquired so far is split into training and testing data on a 4 to 1 ratio. Following this random augmentation techniques such as rotation, shifting, zooming, and horizontal flipping are applied to the training data to enhance model generation.

The MobileNetV2 is used as the base feature extractor, the model's original fully connected layers are excluded to allow for the addition of custom classification layers. Three dense layers with ReLU activation function are added to the model with 512, 256, and 128 neurons respectively. The input images to the model are resized to 224x224x3 to fit the requirements of the base model. Finally, the output layer was modified to be a SoftMax activation function-based dense layer with five target classes.

The model is now trained using the Adam optimizer and categorical cross entropy as the loss function. The model is trained for 30 epochs and continuous validation accuracy was measured with a 20% split in the training data.

D. Prediction and Post-processing

The images are finally passed through the model for prediction of their food classes from the given categories. As we have previously checked for the presence of food within the image inputted, it highly decreases the chance of errors within the model output.
Figure 4 shows the various food items labeled with their predicted classes and outlines the circles of their food items.

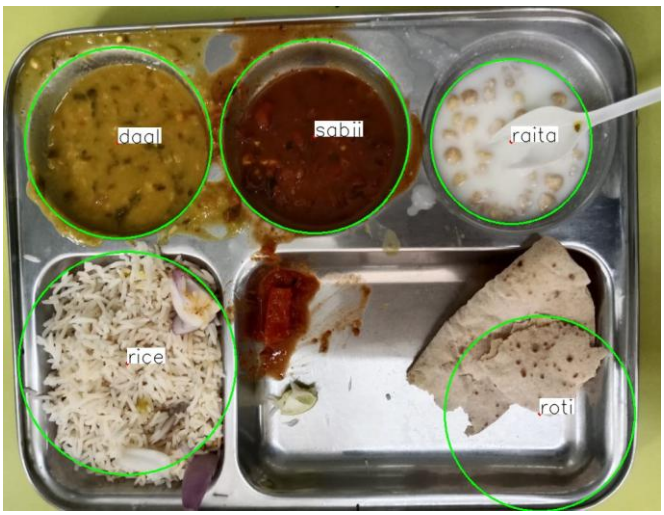


Figure 4 Different Food Items Labelled

Once the various data regarding the food items, and the weight is collected it is stored with a UID for better management. This data is then enriched with metadata such as timestamps, coordinates, etc. In *Figure 5* the entire processing of the food detection system is described, and each major step along the image's journey into the conversion to helpful information has been shown.

For extra scrutiny and better model training for the future, the cropped images are also stored and preferably augmented using techniques such as rotation, scaling, and flipping to increase the diversity of the training dataset. This helps improve the robustness of the machine-learning model.

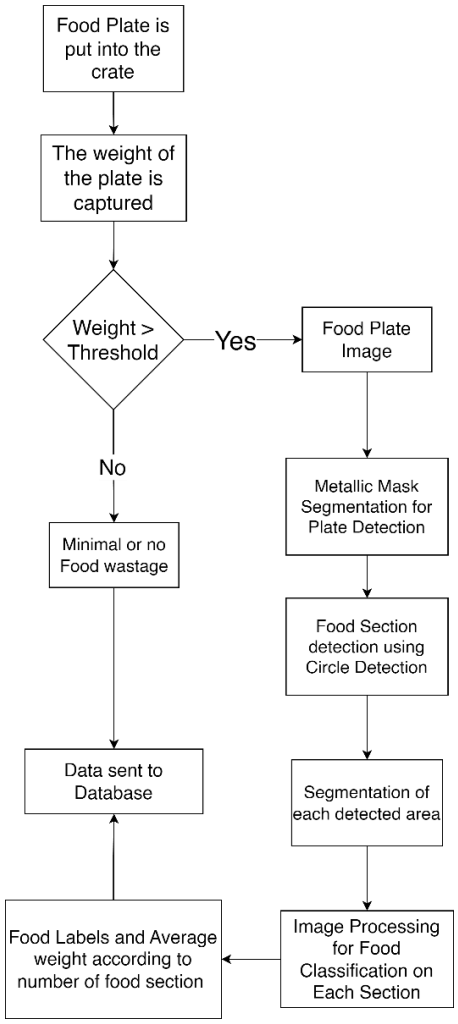


Figure 5 Flow of the Algorithm

IV. Experimental Setup

In this section, we will discuss the particulars of the experimentation that occurred for this technique.

A. Database

The databases used for the experimentation were collected by the team manually. Various dishes' images were captured using the same experimental setup. The data contains images from a wide variety of locations and different kinds of dishes. Table 3 shows the value of different classes and their sample size.

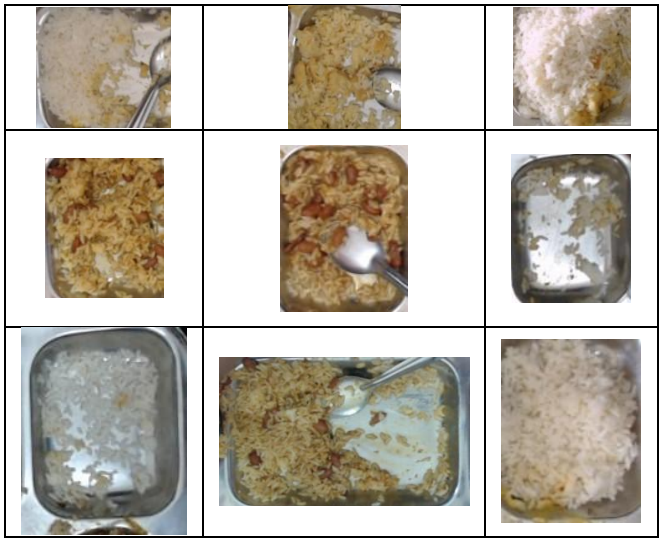
Table 3 Description of the Database

S. No.	Class Name	Sample Size
1.	Roti	82
2.	Rice	78
3.	Sabji (Vegetable Dish)	87
4.	Raita (Curd)	75
5.	Daal	79

Table 3 and Table 4 show some images from the Sabji and Rice classes of the database respectively. This showcases the various types of images with different orientations, sizes, quality, and textures that exist within the database.

Table 4 Eggplant Dish Wastage Images

Table 5 Rice Wastage Images



This data was collected and then pre-processed to better train the model. Using the steps discussed previously to detect the metal plate and automatically crop out the various sections of the plate with different dishes.

B. Evaluation Metrics

In this work, we use established assessment metrics, such as Accuracy, Precision, Recall, and the Area Under the Curve (AUC), to evaluate the data processing technique and the resulting machine learning models. By measuring the model's capacity for accurate prediction, these metrics offer a thorough grasp of model performance, especially in classification tasks.

1) Accuracy

Accuracy measures the proportion of correctly classified instances out of the total number of instances. It is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

2) Precision

Precision quantifies the proportion of correctly predicted positive instances among all instances classified as positive. It is particularly useful in applications where minimizing false positives is crucial. The formula is given by:

$$Precision = \frac{TP}{TP + FP}$$

3) Recall

Recall measures the proportion of actual positive instances that are correctly identified by the model. It is especially important in scenarios where minimizing false negatives is critical. The formula is:

$$Recall = \frac{TP}{TP + FN}$$

4) Area Under Curve

The Receiver Operating Characteristic (ROC) curve is a graphical representation of a model's classification performance across different threshold values. The Area Under the Curve (AUC) quantifies this performance, with values ranging from 0 to 1:

$$AUC = \int_0^1 TPR(FPR)d(FPR)$$

Where:

- **True Positive Rate (TPR) = Recall**
- **False Positive Rate (FPR)** is defined as:

$$FPR = \frac{FP}{FP + TN}$$

Where:

- TP (True Positives) are correctly predicted positive instances.
- TN (True Negatives) are correctly predicted negative instances.
- FP (False Positives) are incorrectly predicted positive instances (Type I error).
- FN (False Negatives) are incorrectly predicted negative instances (Type II error).

V. Results and Discussion

In the below *Table 6* we show the results of the model based on the evaluation metrics.

Table 6 Various Metrics and their Values

Metric	Value
Loss	0.4823
Accuracy	0.8846
Precision	0.8212
Recall	0.8153
Area Under Curve	0.9644

The model was made to predict 4 distinct classes of food that are found commonly in communal eating spaces. A confusion matrix of which can be found in *Figure 6*. The table shows the perfect result

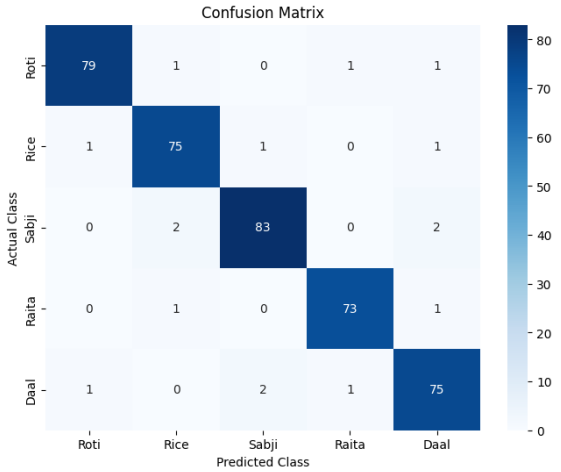


Figure 6 Confusion Matrix of Model

In *Figure 7* we compare various metrics such as Precision, Recall, and F1-Score from various other works that deal with the use of computer vision models for food detection.

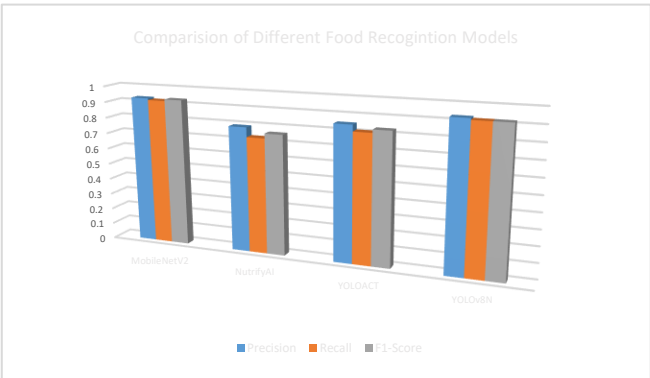


Figure 7 Comparison between various models

We first look at the work on the NutrifyAI by Han M et al where they work on introducing advanced computer vision to detect food items and analyze their nutritional values.[12] Moreover, we look at work by Dai X. who in their studies to detect food in smart refrigerators using the variations of the YOLO model.[13] However, we can say that our model with the MobileNetV2 along with specialized pre-processing results in equal or better metrics compared to these various studies.

During our research, we implemented the system in the institution hostel and followed the trend of food wastage over the course of two weeks. We analyzed that on day 1 before the implementation of the system, an estimated 7 kg of food wastage occurred during the lunch serving. After the proposed system was implemented, and the data from the system was sent to the concerned authorities the serving sizes of the most affected food items were decreased and special care was put into preparing them.

Figure 8 represents the amount of food each class wasted on a certain day, for 7 continuous days the food wastage seemed to drastically decrease however on the 8th day it came to a halt reaching a saturation point, from here on out for the next week of observation no major changes were observed in the reduction of food wastage.

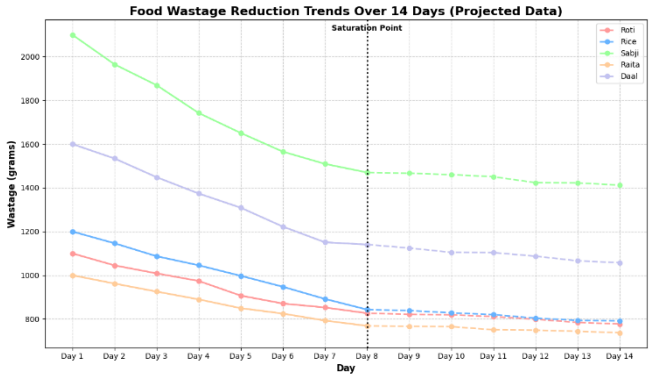


Figure 8 Food wastage analysis

VI. Conclusion and Future Work

The system, which is designed for easy implementation across multiple locations, can also be mounted and interfaced with robotic devices to significantly enhance its functionality. By integrating this system with a mobile robotic device, it is possible to create a smart and mobile dustbin suitable for use at events and within organizations. This setup allows diners to dispose of their food plates conveniently while simultaneously enabling continuous analysis of food wastage.

- The smart and mobile dustbin autonomously navigates the venue, collecting food waste efficiently.
- Its mobility ensures that it can reach diners wherever they are, improving convenience and encouraging proper waste disposal.
- Sensors and AI-driven technology within the system can categorize and sort the waste, ensuring accurate data collection.

As diners dispose of their food plates, in a system like the one shown in *Figure 7*. The system continuously monitors and analyzes the waste. This real-time data collection provides valuable insights into consumption patterns and wastage trends. The data collected by the system can be aggregated and analyzed to inform long-term strategies for food

Figure 7 Implementation of a Computer Vision based Food Detection Bin

management. Trends and patterns observed over time can guide future menu planning, inventory management, and waste reduction initiatives. This data-driven approach ensures that decisions are based on accurate, real-time information, leading to more effective outcomes.

Moreover, mobile device-based applications can be created that can directly access the data stored in the cloud in real-time to give insights about the food wastage occurring at the various venues to the concerned people making them aware of the wastage and thus in turn allowing them to take steps to reduce the wastage. Table 7 shows an example UI of a mobile-based application that might be created to provide real-time insights into the food wastage situation of an organization.

Table 7 Example UI for the application

The below cell contains a visual representation of how data collected by the system can be displayed.

The below cell shows how visual graphs can be created to better visualize food wastage and its effects.

The below cell shows individual plates and their food wastage data.

VII. References

[1] M. Kummu, H. de Moel, M. Porkka, S. Siebert, O. Varis, and P. J. Ward, "Lost food, wasted resources: Global food supply chain losses and their impacts on freshwater, cropland, and fertilizer use," *Science of the Total Environment*, vol. 438, pp. 477–489, Nov. 2012, doi: 10.1016/j.scitotenv.2012.08.092.

[2] C. H. Huang, S. M. Liu, and N. Y. Hsu, "Understanding global food surplus and food waste to tackle economic and environmental sustainability," *Sustainability (Switzerland)*, vol. 12, no. 7, Apr. 2020, doi: 10.3390/su12072892.

[3] S. Gull, I. S. Bajwa, W. Anwar, and R. Rashid, "Smart eNose Food Waste Management System," *J Sens*, vol. 2021, 2021, doi: 10.1155/2021/9931228.

[4] Y. Chauhan, "Food waste management with technological platforms: Evidence from Indian food supply chains," *Sustainability (Switzerland)*, vol. 12, no. 19, Oct. 2020, doi: 10.3390/su12198162.

[5] M. Faezirad, A. Pooya, Z. Naji-Azimi, and M. Amir Haeri, "Preventing food waste in subsidy-based university dining systems: An artificial neural network-aided model under uncertainty," *Waste Management and Research*, vol. 39, no. 8, pp. 1027–1038, Aug. 2021, doi: 10.1177/0734242X211017974.

[6] U. Kumar and Y. Kumar, "Food Waste Management Android App." [Online]. Available: https://ssrn.com/abstract=4157538

[7] J. Bharucha, "Tackling the challenges of reducing and managing food waste in Mumbai restaurants," *British Food Journal*, vol. 120, no. 3, pp. 639–649, 2018, doi: 10.1108/BFJ-06-2017-0324.

[8] V. Filimonau, V. N. Nghiem, and L. Wang, "Food waste management in ethnic food restaurants," *Int J Hosp Manag*, vol. 92, p. 102731, Jan. 2021, doi: 10.1016/j.ijhm.2020.102731.

[9] A. Saadat, S. Barman, and L. I. Shukla, "Reduction of Food Wastage: Roles of Administrative Interventions

and Cultural Ecosystem Services," 2020, pp. 671–691. doi: 10.1007/978-3-319-95675-6_97.

[10] A. Sahoo, A. Dwivedi, P. Madheshiya, U. Kumar, R. K. Sharma, and S. Tiwari, "Insights into the management of food waste in developing countries: with special reference to India," *Environmental Science and Pollution Research*, vol. 31, no. 12, pp. 17887–17913, Mar. 2024, doi: 10.1007/s11356-023-27901-6.

[11] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," Jan. 2018, [Online]. Available: http://arxiv.org/abs/1801.04381

[12] M. Han, J. Chen, and Z. Zhou, "NutrifyAI: An AI-Powered System for Real-Time Food Detection, Nutritional Analysis, and Personalized Meal Recommendations," Aug. 2024, [Online]. Available: http://arxiv.org/abs/2408.10532

[13] X. Dai, "Robust deep-learning based refrigerator food recognition," *Front Artif Intell*, vol. 7, 2024, doi: 10.3389/frai.2024.1442948.